Trauma

STB, Spinal Immobilization, and Splinting

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Scenario 1: 22 YOM fall from building

What should you treat first after arriving on scene? How?

Exsanguinating hemorrhage, tourniquet

Assessing Trauma

XABCD

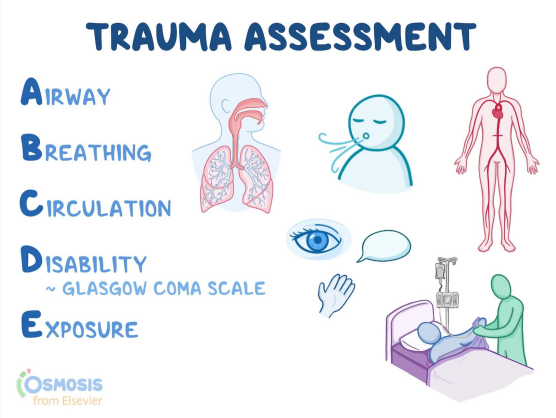
- **X**: life threats
- **A**: airway
- **B**: breathing
- **C**: circulation
- **D**: disability/transport decision

Trauma Mechanisms

- Axial loading
- Blunt and penetrating trauma
- Collisions

Rapid head to toe assessment

Look for DCAP-BTLS





Spinal Motion Restriction

What does SMR consist of?

- Reduce gross motion by pt
- Prevent duplicating damaging mechanism to spine
- Regularly reassess PMS/CMS

Consider SMR for patients who may have traumatic unstable spinal column injury

SMR methods/tools include:

- Cervical collar with positioning
- KED
- Backboard with adequate padding (hard backboard utilization should be limited)
- Head immobilizer and straps



Bleeding Management

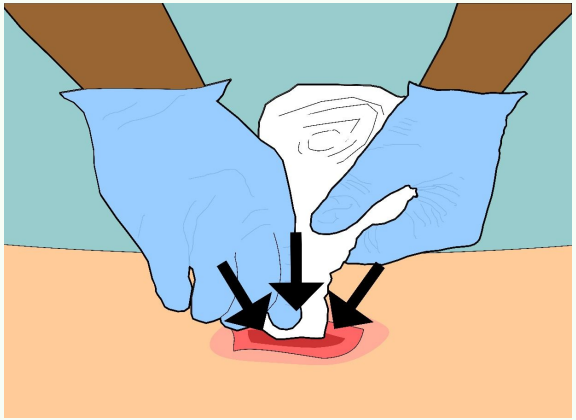
Tourniquet

- Uncontrollable bleeding in the extremities
- Apply 2–3 in above wound/high and tight

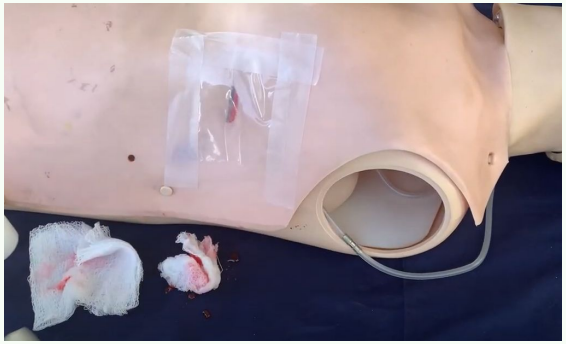


Wound packing

- Uncontrolled bleeding from extremity and junctional wounds
  - May use standard or hemostatic gauze



Bleeding Management (cont.)





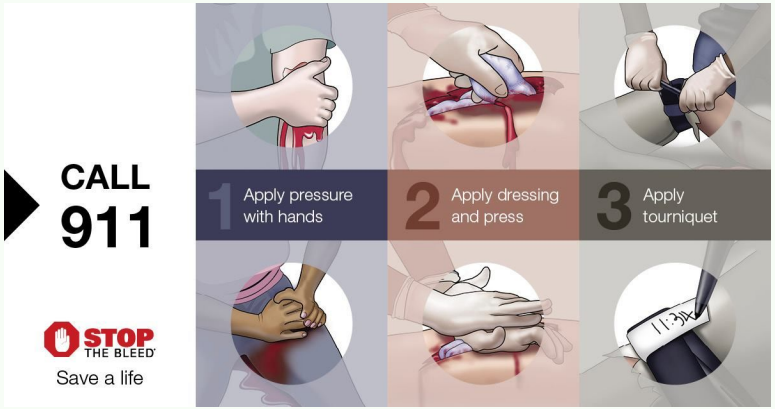
Occlusive dressing

- Sealing wounds, preventing air entry/fluid loss
    - Sucking chest wound, abdominal evisceration, neck wounds

Hemostatic gauze

- After tourniquet placement, and to aid in severe arterial bleeding
- Control severe bleeding where tourniquets are not indicated (but NOT in internal cavities)
- Use is optional

Stop the Bleed

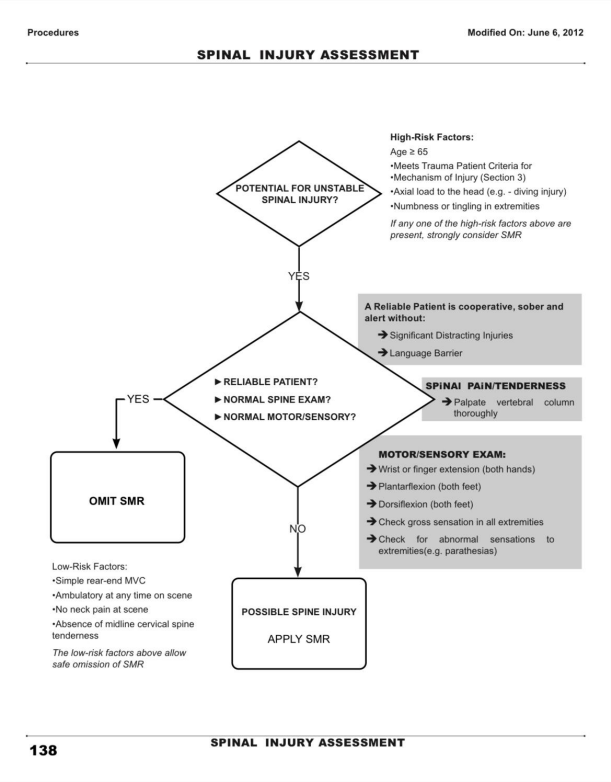


Personal safety

ABCs of bleeding control

- **A**: Alert 911
- **B**: Locate Bleeding
- **C**: Compress (**3 Ps**)
  - Pressure
  - Packing
  - Putting on a tourniquet

ALCO Protocol



Procedures

Modified On: June 6, 2012

SPINAL INJURY ASSESSMENT

POTENTIAL FOR UNSTABLE SPINAL INJURY?

High-Risk Factors:
Age ≥ 65
•Meets Trauma Patient Criteria for •Mechanism of Injury (Section 3)
•Axial load to the head (e.g. - diving injury)
•Numbness or tingling in extremities

If any one of the high-risk factors above are present, strongly consider SMR

YES

A Reliable Patient is cooperative, sober and alert without:
→ Significant Distracting Injuries
→ Language Barrier

► RELIABLE PATIENT?
► NORMAL SPINE EXAM?
► NORMAL MOTOR/SENSORY?

YES

SPINAL PAIN/TENDERNESS
→ Palpate vertebral column thoroughly

MOTOR/SENSORY EXAM:
→ Wrist or finger extension (both hands)
→ Plantarflexion (both feet)
→ Dorsiflexion (both feet)
→ Check gross sensation in all extremities
→ Check for abnormal sensations to extremities(e.g. parathesias)

OMIT SMR

NO

POSSIBLE SPINE INJURY

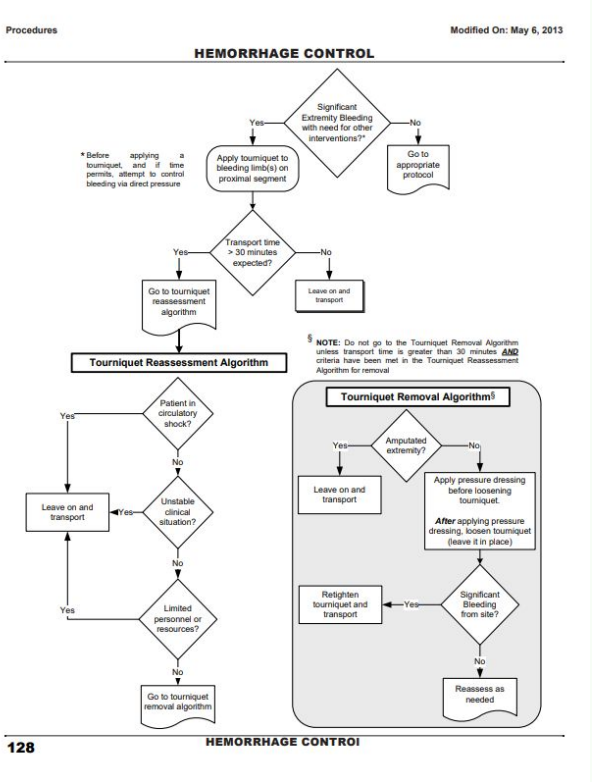
APPLY SMR

Low-Risk Factors:
•Simple rear-end MVC
•Ambulatory at any time on scene
•No neck pain at scene
•Absence of midline cervical spine tenderness

The low-risk factors above allow safe omission of SMR

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SPINAL INJURY ASSESSMENT



Procedures

Modified On: May 6, 2013

HEMORRHAGE CONTROL

Significant Extremity Bleeding with need for other interventions?*

Yes

No

* Before applying a tourniquet, and if time permits, attempt to control bleeding via direct pressure

Apply tourniquet to bleeding limb(s) on proximal segment

Go to appropriate protocol

Transport time > 30 minutes expected?

Yes

No

Go to tourniquet reassessment algorithm

Leave on and transport

Tourniquet Reassessment Algorithm

§ NOTE: Do not go to the Tourniquet Removal Algorithm unless transport time is greater than 30 minutes ***AND*** criteria have been met in the Tourniquet Reassessment Algorithm for removal

Patient in circulatory shock?

Yes

No

Leave on and transport

Yes

Unstable clinical situation?

No

Yes

Limited personnel or resources?

No

Go to tourniquet removal algorithm

Tourniquet Removal Algorithm§

Amputated extremity?

Yes

No

Leave on and transport

Apply pressure dressing before loosening tourniquet.

After applying pressure dressing, loosen tourniquet (leave it in place)

Retighten tourniquet and transport

Yes

Significant Bleeding from site?

No

Reassess as needed

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HEMORRHAGE CONTROL

What is your transport decision?

Code 3
Trauma Center

Defining Trauma Severity

Red (High)

Rapid transport to trauma center

Injury patterns

- Penetrating injury to head, neck, torso, proximal extremities
- Skull deformity, fracture
- Spinal injury
- Chest wall
- Active bleeding
- Fracture 2+ long bones
- Crushed, degloved, mangled, pulseless extremity
- Amputation proximal to wrist/ankle

Mental Status/Vitals (general)

- Total GCS <13 or motor <6
- RR <10 or >29
- Respiratory distress/need for support
- Room air spO2 <90%

Defining Trauma Severity (cont.)

MOI

- High-risk auto crash
- Rider separated from vehicle with significant impact
- Pedestrian/biker thrown, run over with significant impact
- Fall from height >10 ft

Considerations include: age, pregnancy, anticoagulant use, burns, medical history

Yellow (Moderate)

Generally transport C2 (use judgment) to trauma center

ALCO Protocol

TRAUMA PATIENT CRITERIA

1. **INTRODUCTION:** The goal of the Alameda County trauma system is to transport confirmed patients meeting the various criteria below to a designated trauma center in a timely manner, bypassing non-trauma centers

2. **RED CRITERIA TRAUMA PATIENTS (High Risk for Serious Injury):**

   2.1 A patient is identified as at high risk for serious injury when any of the following injury patterns or mental status/vitals signs listed below are present. These patients should be transported to a designated Trauma Center rapidly.

Injury Patterns	Mental Status & Vitals Signs
• Penetrating injuries to head, neck, torso,and proximal extremities • Skull deformity, suspected skull fracture • Suspected spinal injury with new motor or sensory loss • Chest wall instability, deformity, or suspected flail chest • Suspected pelvic fracture • Suspected fracture of two or more proximal long bones • Crushed, degloved, mangled, or pulseless extremity • Amputation proximal to wrist or ankle • Active bleeding requiring a tourniquet or wound packing with continuous pressure	**All Patients** • Total Glasgow Coma Scale ≤ 13 ***or*** Motor GCS < 6 (Unable to follow commands) • RR < 10 or > 29 breaths/min • Respiratory distress or need for respiratory support • Room-air pulse oximetry < 90% **Age 0–9 years** • SBP < 70mm Hg + (2 x age in years) **Age 10–64 years** • SBP < 90 mmHg or • HR > SBP **Age ≥ 65 years** • SBP < 110 mmHg or • HR > SBP

3. **YELLOW CRITERIA TRAUMA PATIENTS (Moderate Risk for Serious Injury):**

   3.1 In addition to above criteria, the following mechanisms of injury and EMS provider judgment of risk factors can be utilized to preferentially triage a patient to a trauma center. In general, these patients are transported code 2, however, differing field circumstances and/or patient condition may require a code 3 transport

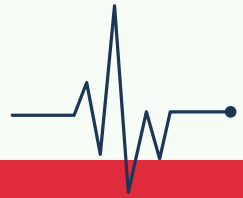
Mechanism of Injury	EMS Judgment
• High-Risk Auto Crash – Partial or complete ejection – Significant intrusion (including roof) • >12 inches occupant site OR • >18 inches any site OR • Need for extrication for entrapped patient – Death in passenger compartment – Child (age 0–9 years) unrestrained or in unsecured child safety seat – Vehicle telemetry data consistent with severe injury • Rider separated from transport vehicle with significant impact (eg, motorcycle, ATV, horse, etc.) • Pedestrian/bicycle rider thrown, run over, or with significant impact • Fall from height > 10 feet (all ages)	**Consider risk factors, including:** • Low-level falls in young children (age ≤ 5 years) or older adult (age ≥ 65 years) with significant head impact • Anticoagulant use • Suspicion of child abuse • Special, high-resource healthcare needs • Pregnancy > 20 weeks • Burns in conjunction with trauma • Children should be triaged preferentially to pediatric capable centers • EMS Provider judgment - If concerned, take to a trauma center

Considerations for this patient

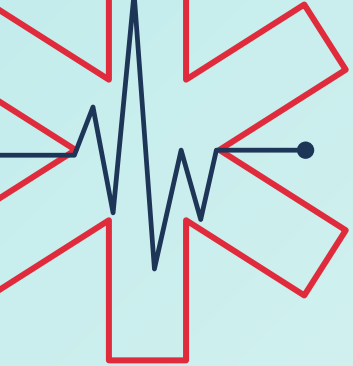
- Scene safety
- Immediately stop the bleed, put on tourniquet
- C spine, SMR
- Rapid transport to trauma center
- Splint deformed leg
- Apply gauze to head abrasion
- Keep pt warm, supine

BMRC First Aid Protocol

- Call 911 or UCPD
  - Be ready to give report
- Apply tourniquet
- Measure vital signs and closely monitor every 5 minutes
- If higher level of care provider present, may splint



Scenario 2: 15 Y/O M in a tractor collision



What is your field impression?

Hemorrhagic and hypovolemic shock

Ortho Injuries

Fractures & Breaks in Extremities

- Closed vs. Open
- Common Types:
  - Greenstick
  - Transverse
  - Oblique
  - Spiral
  - Comminuted

Pelvic Girdle Fractures

- Causes the most significant bleeding (major blood vessels running through this area)

Dislocations

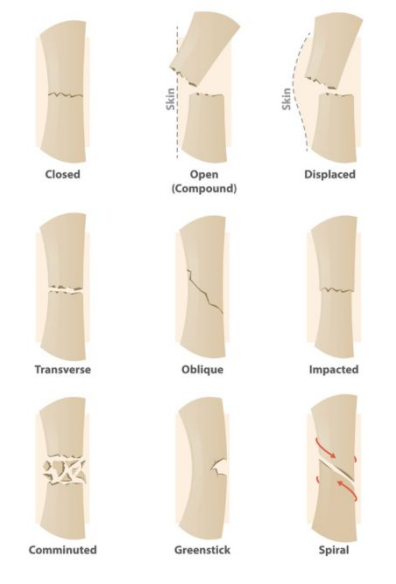
- Common sites: joint connecting collarbone to shoulder, the shoulder itself, elbows, wrist, fingers, hip, and knee

Bone bruises

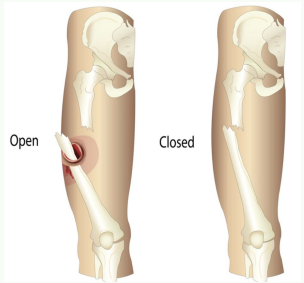
- Traumatic micro-injury when small small blood vessels bleed into bone tissue, causing deep pain and swelling

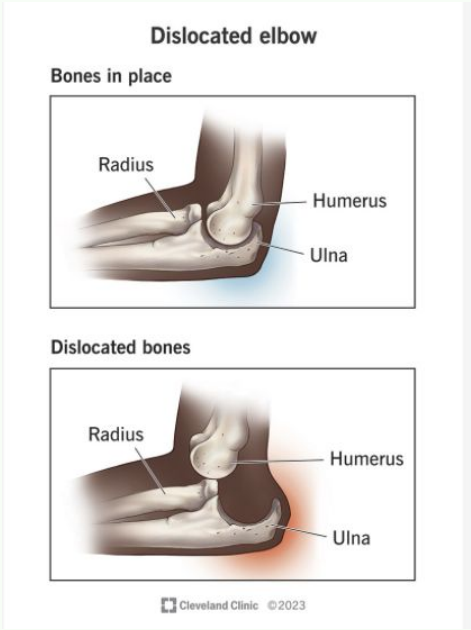


Ortho Injuries (cont.)









What two types of fractures are the most life-threatening?

Answer: Pelvic and femoral

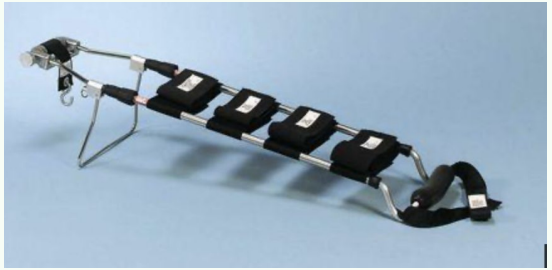
Ortho Injuries: Splinting

- First focus on hemorrhage control (if applicable)
- Then immobilization + splinting *check CSM before + after applying*
  - Rigid splint
  - Soft splint
  - Vacuum splint
  - Traction splint (if suspected femoral shaft fracture)
- Things to remember:
  - Remove/cut clothing when needed to assess injury
  - Do not force a deformed extremity into normal alignment
  - Monitor for shock + transport promptly when indicated









ALCO Protocol

EXTREMITY INJURY

1. **ASSESMENT:**
   - 1.1. Routine Medical Care
   - 1.2. Document mechanism of injury
   - 1.3. Document past medical history including history of previous injuries
   - 1.4. Check for deformity, open wounds, swelling, shortening, and/or rotation
   - 1.5. Document range of motion, pulses, sensation, and color of the extremity
   - 1.6. Assess severity of pain (1-10 scale)
   - 1.7. Assess for other associated injuries
2. **GENERAL CARE:** (all patients)
   - 2.1. Control any external bleeding with direct pressure
   - 2.2. Elevate and apply cold packs
   - 2.3. Splint injured extremity. Hand injuries should be positioned in the "safe position"
   - 2.4. Cover open wounds with sterile dressings
   - 2.5. Provide Pain control – see "Pain Management" **page 47** (Adult) or **page 76** (Pediatric)
   - 2.6. Remove rings or other possibly constricting items



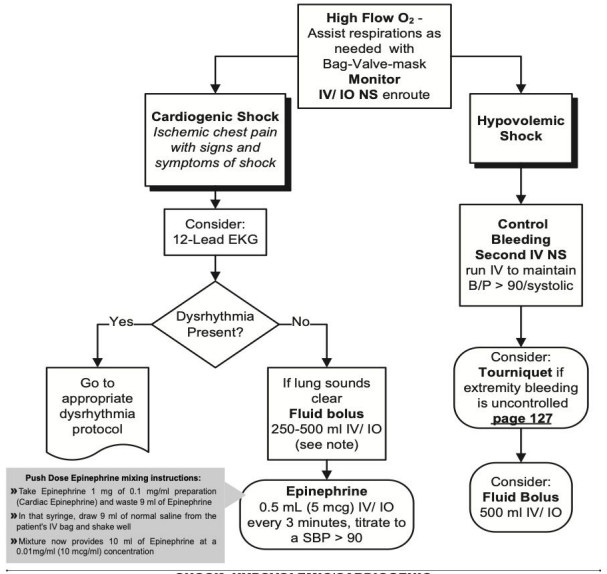
Safe Position

3. **FRACTURE/DISLOCATION:**
   - 3.1. If the extremity is pulseless, attempt to place it in normal anatomic position by gentle in-line traction
   - 3.2. If repositioning does not restore circulation, do not manipulate further, transport immediately.
   - 3.3. Start IV NS in uninjured extremity

(for ortho extremity injury)

SHOCK: HYPOVOLEMIC/CARDIOGENIC

- **Routine Medical Care**
- **Shock** - 2 or more of the following:
  - → Pulse > 120/minute
  - → BP < 90/systolic
  - → Altered Mental Status
  - → Pale, cool and/or diaphoretic skin signs
- Initiate early transport and treat en route, if appropriate.
- **NOTE:** A fluid bolus of up to 500 ml Normal Saline may be given to an adult patient in cardiogenic shock with clear lung sounds.
- If **anaphylaxis suspected**, see **page 36**
- If **trauma suspected**, see **page 28**
- If **sepsis suspected**, see **page 56**



(for shock)

Types of Skin Wounds

DEFINITIONS:



Abrasion
- Superficial scrape of skin
- Ex: road rash, scraped knee

Laceration
- Cut/tear in the skin

Evisceration
- Internal organs protrude through open wound in the abdominal wall

Avulsion
- Skin or tissue is torn away

Contusion
- Bruising caused by a blunt force
- Blood vessels are damaged under intact skin

Types of Skin Wounds

DEFINITIONS:	TREATMENTS:
Abrasion • Superficial scrape of skin • Ex: road rash, scraped knee	Apply direct pressure w/ sterile gauze.
Laceration • Cut/tear in the skin	Direct pressure → wound packing → tourniquet.
Evisceration • Internal organs protrude through open wound in the abdominal wall	Do not push organs back in. Apply moist, sterile dressing + occlusive dressing. Control bleeding only on surrounding edges.
Avulsion • Skin or tissue is torn away	Direct pressure → tourniquet. Gently fold skin flap back into anatomical position.
Contusion • Bruising caused by a blunt force • Blood vessels are damaged under intact skin	Towel – wrapped cold pack + light compression. Watch for compartment syndrome.

Scenario 3: 32 Y/O M in High-Speed MVC

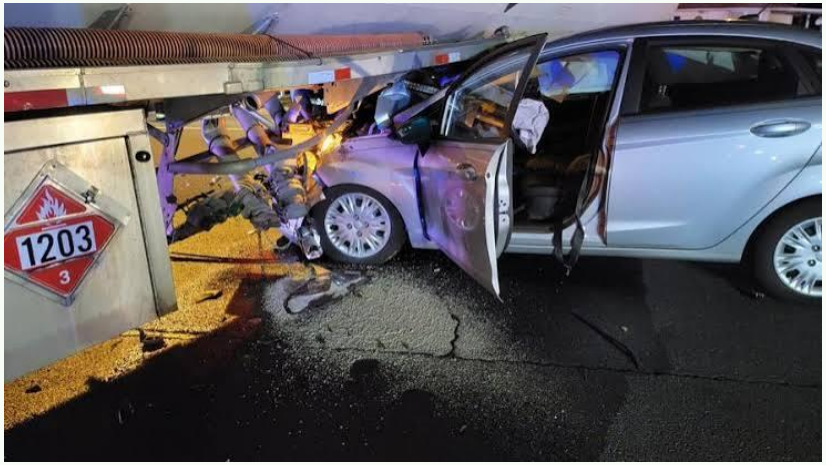
What injuries did the patient sustain?

Answer:

Flail chest,
coup–contrecoup







SCENE SAFETY

- Ensure that traffic is controlled
- Look for any other hazards due to the accidents:
  - Downed power lines
  - Shattered glass
  - Fuel leakage/unidentified liquid
  - Fire concerns
  - Unstable vehicle
  - Airbags
  - etc.





SCENE SAFETY

- Park ambulance in appropriate location depending on environment
- Ensure it can leave the scene easily
- Lights on
- What precautions should EMS take when called to road accidents at night?





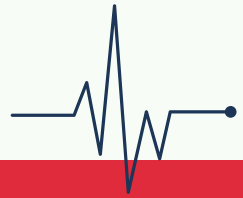
SCENE SAFETY

- Park ambulance in appropriate location depending on environment
- Ensure it can leave the scene easily
- Lights on
- What precautions should EMS take when called to road accidents at night?

→ Wear high-visibility: reflective safety vest or approved high-visibility outerwear

Primary assessment for this patient

A: How do you check airway?



Primary assessment for this patient

A: How do you check airway?

→ JAW THRUST MANEUVER



Primary assessment for this patient

- **A**: How do you check airway?
  → JAW THRUST MANEUVER
- **B**: What was the most concerning finding?
- **C**: Poor circulation
- **D**: Transport decision

Trauma Mechanism

- Collision

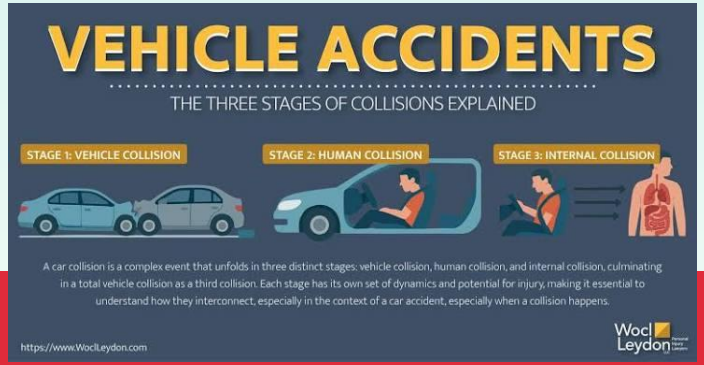
PEARRL

Pupils asymmetrical



The Three Collisions

Every MVC involves three distinct collisions that happen in rapid sequence. **Important to understanding MOI.**



First Collision

VEHICLE VS OBJECT

The car strikes another vehicle or stationary object (e.g. tree, guardrail, etc.)

- Causes vehicle frame to deform (eg. significant front-end intrusion)
- Deceleration



First Collision

VEHICLE VS OBJECT

The car strikes another vehicle or stationary object (e.g. tree, guardrail, etc.)

- Causes vehicle frame to deform (eg. significant front-end intrusion)
- Deceleration

- Vehicle struck concrete barrier



Second Collision

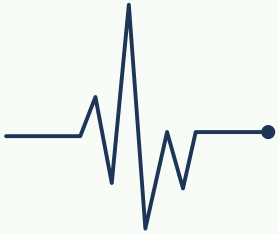
PASSENGER VS INTERIOR

The passenger(s) collide with objects within vehicle (traveling at the vehicle's original speed)

- Steering wheel, dashboard, windshield, seatbelts, other interior structures.



Second Collision



PASSENGER VS INTERIOR

The passenger(s) collide with objects within vehicle (traveling at the vehicle's original speed)

- Steering wheel, dashboard, windshield, seatbelts, other interior structures.

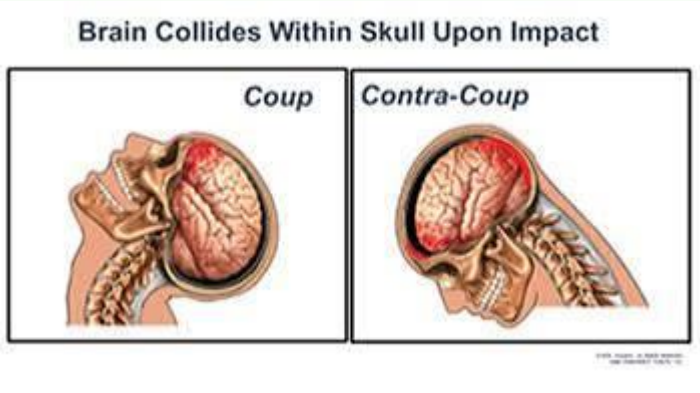
- Damage to steering wheel → rib fractures
- Windshield spider webbing → potentially head was struck, but could also be caused by another object

Third Collision

INTERNAL ORGANS VS SKELETON/CAVITIES

The body's internal organs and tissues keep moving forward and strike the rigid skeletal structure

- The skull, rib cage, pelvic wall
- Tear against supporting ligaments and blood vessels
    - aortic rupture



Third Collision

INTERNAL ORGANS VS SKELETON/CAVITIES

The body's internal organs and tissues keep moving forward and strike the rigid skeletal structure

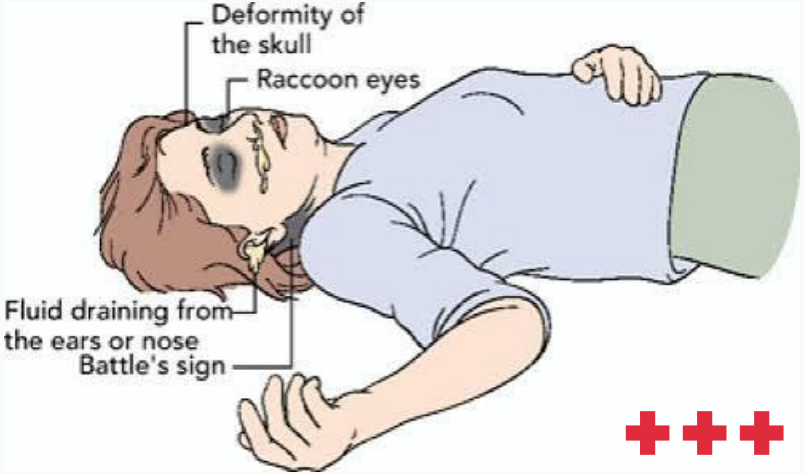
- The skull, rib cage, pelvic wall
- Tear against supporting ligaments and blood vessels
    - aortic rupture

- Coup-contrecoup
    - Coup injury: damage occurs directly under the site of the impact where the brain hits the skull
    - Contrecoup injury: damage occurs on exact opposite side of the brain, when brain rebounds against opposite side of the skull
    - Coup-Contrecoup injury: combination of both injuries from single, severe blow, car crash, or heavy fall



TBI/Neurological Considerations

- AVPU, A&O, GCS
- PEARRL
  - Anisocoria
- BE FAST
- CSM
- Vital signs (BGL, pulse ox, temperature)
  - Low blood sugar, hypoxia mimic stroke and seizures
  - Cushing's Triad (widening pulse pressure, bradycardia, irregular respirations)
- TBI SECONDARY INJURY
  - hypoxia, hypotension, and hyperventilation
- Palpation and visual check
  - Racoon eyes, battle signs, CSF



Overall: Necessary interventions for this patient

- Spinal immobilization
- Supplemental O_2 (NRB or BVM depending on breathing adequacy)
- Keep patient warm (treating for shock)
- Ongoing, frequent reassessment for critical patient, including neurological monitoring
- ALS
- **RAPID TRANSPORT**

ALCO PROTOCOL

RED CRITERIA TRAUMA PATIENTS (High Risk for Serious Injury):

- Chest wall instability, deformity, or suspected flail chest

All Patients

- Total Glasgow Coma Scale ≤ 13 or; Motor GCS < 6 (Unable to follow commands)
- RR < 10 or > 29 breaths/min
- Respiratory distress or need for respiratory support
- Room-air pulse oximetry < 90%

ALCO PROTOCOL

- **Care of the patient with a closed head injury** (GCS < 8):
  - → **Advanced airway (ETT or SGA)**
  - → End-tidal CO_2 should be between 30-35 mmHg
  - → Track respirations or ventilate to a rate of approx 12 times/minute with 100% O_2 ***(AVOID HYPERVENTILATION)***
  - → **IV/IO** NS in 500 mL increments to ***maintain mean arterial pressure (MAP) of at least 80 mmHg***. Reassess BP q 5 minutes

MPORTANT CONSIDERATIONS

- Contact the Base Hospital, if appropriate
- Contact the Trauma Center, as soon as possible
- Consider pain management when appropriate
- Splint fractures and dress wounds ONLY if time permits

FORMULA FOR ESTIMATING MAP

$$\text{MAP} = \text{diastolic} + \frac{(\text{systolic} - \text{diastolic})}{3}$$

6. **PATIENT TURNOVER REPORTING FORMAT:** EMS Clinicians should use the following **DMIST** format when turning over patient care to the Trauma Center medical team:
   - → **D**emographics
   - → **M**echanism
   - → **I**njuries
   - → **S**igns - Vital signs
   - → **T**reatments

KAHOOT TIME

Thank you

10/12/2026

